

# Folium, PyWaffle, WordCloud, and Seaborn Cheat Sheet

## 1 Folium

| Feature                        | Description                                                                         | Syntax                                                                                                                                                                                                                                    | Example                                                                                                                                                                                                                            |
|--------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Map</b>                     | Create an interactive map centered at a specified location with a given zoom level. | <code>folium.Map(location=[lat, lon], zoom_start=n)</code>                                                                                                                                                                                | <code>world_map = folium.Map() canada = folium.Map(location=[56.130, -106.35], zoom_start=4)</code>                                                                                                                                |
| <b>Marker (Stamen Toner)</b>   | Add a marker to the map with a custom tooltip or popup.                             | <code>folium.Marker( location=[lat, lon], popup='Marker Popup', tiles='Stamen Toner' ).add_to(map)</code>                                                                                                                                 | <code>folium.Marker( location=[56.130, -106.35], tooltip='Marker', tiles='Stamen Toner' ).add_to(world_map)</code>                                                                                                                 |
| <b>Marker (Stamen Terrain)</b> | Add a marker using the Stamen Terrain tile style.                                   | <code>folium.Marker( location=[lat, lon], popup='Marker Popup', tiles='Stamen Terrain' ).add_to(map)</code>                                                                                                                               | <code>folium.Marker( location=[56.130, -106.35], tooltip='Marker', tiles='Stamen Terrain' ).add_to(world_map)</code>                                                                                                               |
| <b>Circle Marker</b>           | Add a circle to the map with a specified radius, color, and fill opacity.           | <code>folium.features.CircleMarker( location=[lat, lon], radius=n, color='red', fill_opacity=n ).add_to(map)</code>                                                                                                                       | <code>folium.features.CircleMarker( location=[56.130, -106.35], radius=1000, color='red', fill_opacity=0.5 ).add_to(world_map)</code>                                                                                              |
| <b>Choropleth Map</b>          | Create a choropleth map based on a GeoJSON file and a specified data column.        | <code>folium.Choropleth( geo_data='path/to/geojson_file', data=df, columns=['region', 'value_column'], key_on='feature.properties.id', fill_color='YlGnBu', fill_opacity=0.7, line_opacity=0.2, legend_name='Legend' ).add_to(map)</code> | <code>world_map.choropleth( geo_data=world_geo, data=df_can, columns=['Country', 'Total'], key_on='feature.properties.name', fill_color='YlOrRd', fill_opacity=0.7, line_opacity=0.2, legend_name='Immigration to Canada' )</code> |

## 2 PyWaffle

| Feature             | Description                                           | Syntax                                                                            | Example                                                                                                                                                                                                |
|---------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waffle Chart</b> | Create a waffle chart based on values and categories. | <code>plt.figure( FigureClass=Waffle, rows=20, columns=30, values=values )</code> | <code>plt.figure( FigureClass=Waffle, rows=20, columns=30, values=df_dsn['Total'], cmap_name='tab20', legend={ 'labels': label, 'loc': 'lower left', 'bbox_to_anchor': (0, -0.1), 'ncol': 3 } )</code> |

|               |                                   |                                                                           |                                                                                  |
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| <b>Legend</b> | Add a legend to the waffle chart. | <code>waffle_chart.legend(loc='upper left', bbox_to_anchor=(1, 1))</code> | <code>waffle_chart.legend(loc='upper left', bbox_to_anchor=(1, 1))</code>        |
| <b>Title</b>  | Add a title to the waffle chart.  | <code>waffle_chart.set_title('Waffle Chart Title')</code>                 | <code>waffle_chart.set_title('Waffle Chart Title')</code>                        |
| <b>Labels</b> | Add labels to the waffle chart.   | <code>waffle_chart.set_labels(['Label 1', 'Label 2', ...])</code>         | <code>waffle_chart.set_labels(['Category A', 'Category B', 'Category C'])</code> |

### 3 WordCloud

| Feature          | Description                                      | Syntax                                                                                                                                                                   | Example                                                                                                                                                                                                  |
|------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WordCloud</b> | Create a word cloud object based on text data.   | <code>wordcloud = WordCloud().generate(text_data)</code>                                                                                                                 | <code>alice_wc = WordCloud(background_color='white', max_words=2000, mask=alice_mask, stopwords=stopwords )<br/>alice_wc.generate(alice_novel)<br/>plt.imshow(alice_wc, interpolation='bilinear')</code> |
| <b>Generate</b>  | Generate the word cloud from the text data.      | <code>wordcloud.generate(text_data)</code>                                                                                                                               | <code>wordcloud.generate(text_data)</code>                                                                                                                                                               |
| <b>Display</b>   | Display the word cloud using Matplotlib.         | <code>plt.imshow(wordcloud, interpolation='bilinear')</code>                                                                                                             | <code>plt.imshow(wordcloud, interpolation='bilinear')</code>                                                                                                                                             |
| <b>Options</b>   | Customize the font, colors, mask, and stopwords. | <code>wordcloud = WordCloud(font_path='path/to/font_file', background_color='white', colormap='Blues', mask=mask_image, stopwords=stopwords ).generate(text_data)</code> | <code>wordcloud = WordCloud(background_color='white', colormap='Blues', stopwords=stopwords ).generate(text_data)</code>                                                                                 |

### 4 Seaborn

| Plot Type        | Description                                                                                            | Syntax                                                                     | Example                                                          |
|------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>barplot</b>   | Create a bar plot to visualize the relationship between a categorical variable and a numeric variable. | <code>sns.barplot(x='x_variable', y='y_variable', data=dataframe)</code>   | <code>sns.barplot(x='Continent', y='Total', data=df_can1)</code> |
| <b>countplot</b> | Create a count plot to display the frequency of each category in a categorical variable.               | <code>sns.countplot(x='category', data=dataframe)</code>                   | <code>sns.countplot(x='Continent', data=df_can)</code>           |
| <b>regplot</b>   | Create a scatter plot with a linear regression line to visualize the relationship                      | <code>sns.regplot( x='x_variable', y='y_variable', data=dataframe )</code> | <code>sns.regplot(x='year', y='total', data=df_tot)</code>       |

|  |                                   |  |  |
|--|-----------------------------------|--|--|
|  | between two<br>numeric variables. |  |  |
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